

APPENDIX D – REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS

D-1 INTRODUCTION

When the Equipment Room is shared by more than one MR system of the same field strength there is a potential for cross-talk of RF energy between the MR systems. RF cross-talk may cause noise artifacts in images.

The potential for cross-talk exists when the RF transmit cables and equipment of two or more MR systems are located in the same Equipment Room. For example, when one system is transmitting, the other system could be in receive mode and therefore pick up the RF energy being transmitted resulting in a cross-talk scenario.

Note

The potential for cross-talk exists for RF transmit cables and equipment that produces RF that are part of a non-GE MR System of the same field strength.

The following subsections provide requirements for shared Equipment Room design, layout, and installation which reduce the potential for RF cross-talk.

D-2 EQUIPMENT CABINETS RELATIVE LOCATIONS

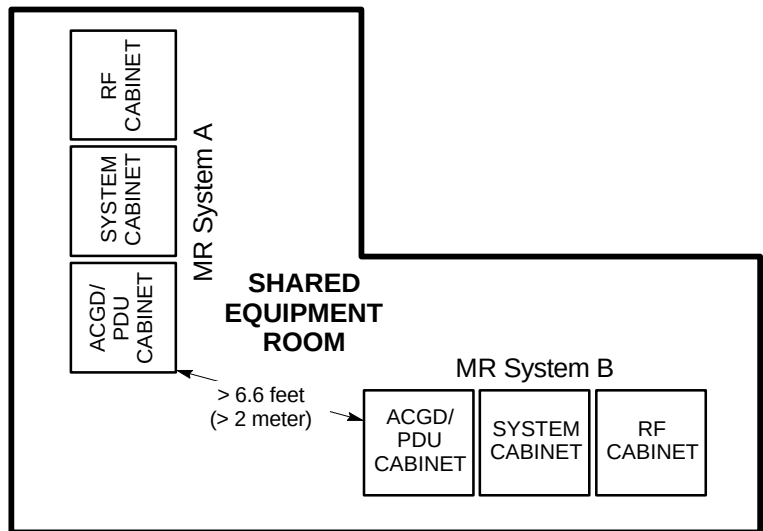
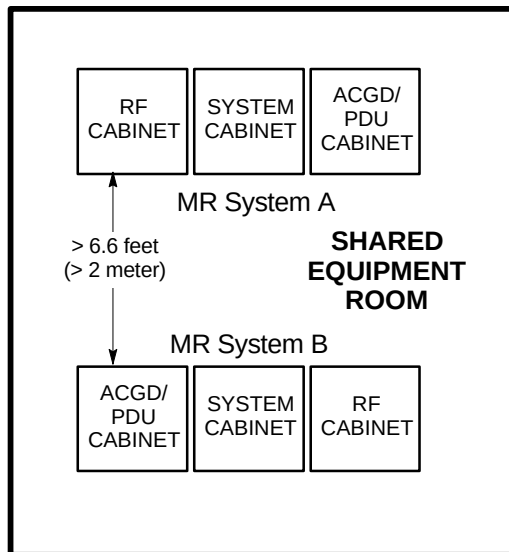
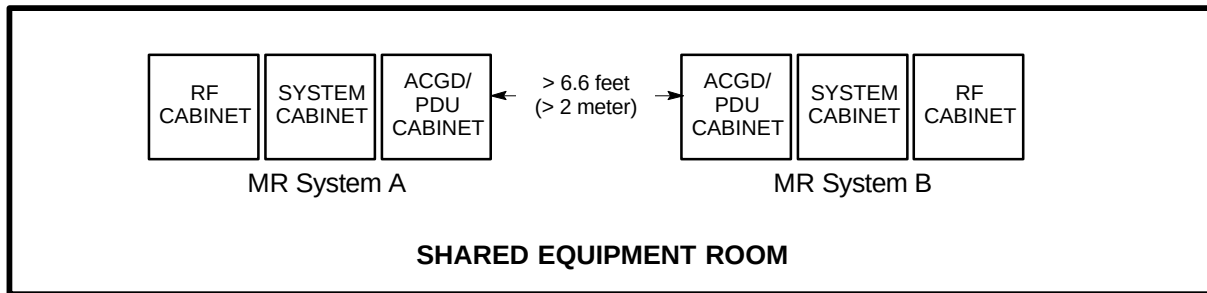
The following are requirements for locating equipment cabinet of one MR system relative to the other MR system equipment cabinets.

- Maximize separation distance between the RF Cabinet of one MR system and the System Cabinet of the other MR system.
- RF, System, or Gradient Cabinets of one MR system and RF, System, or Gradient Cabinets of the other MR system must be separated by a minimum of > 6.6 feet (2 meters) in all directions, see Illustration C-1.

Note

Relative placement should not be an issue for chillers, compressors, and other equipment which does not produce RF.

D-2 EQUIPMENT CABINETS RELATIVE LOCATIONS (Continued)

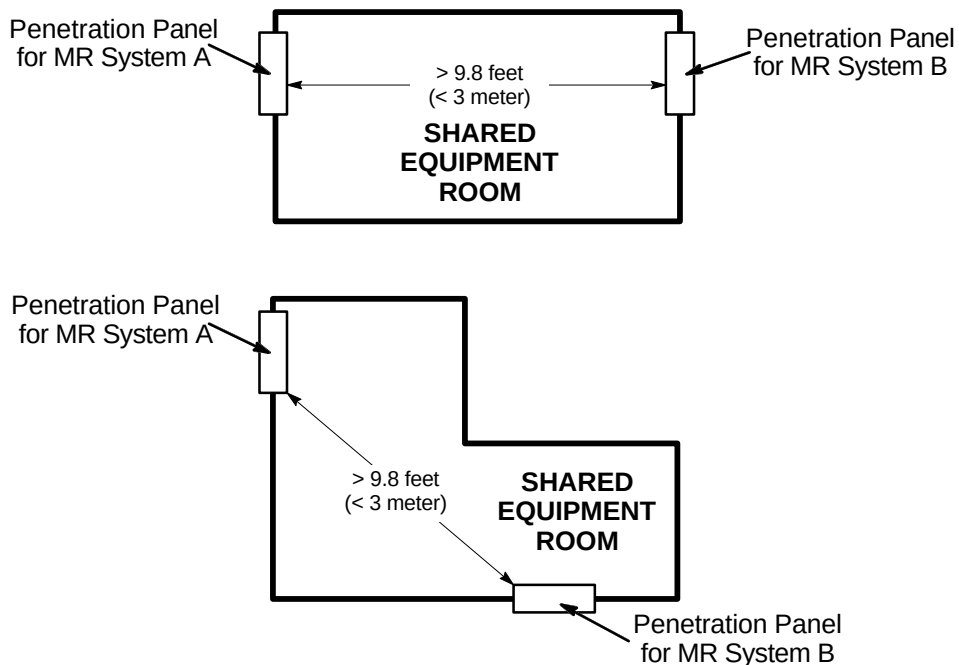


MULTIPLE MR SYSTEMS ELECTRONICS CABINETS SPACING
ILLUSTRATION C-1

D-3 PENETRATION PANELS LOCATION

The following are requirements for locating the RF Shielded Room Penetration Panel of one MR system relative to the other MR system RF Shielded Room Penetration Panel.

- There must be > 9.8 feet (3 meters) separation between the Penetration Panels of each system sharing the Equipment Room space, see Illustration C-2.

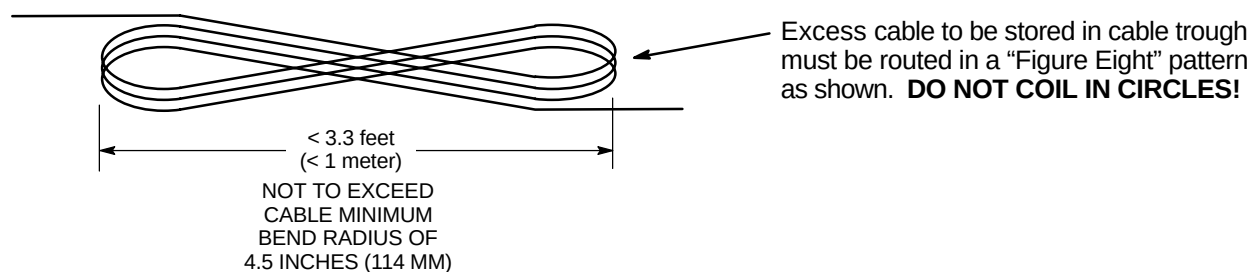


MULTIPLE MR SYSTEMS PENETRATION PANEL SPACING
ILLUSTRATION C-2

D-4 SYSTEM CABLES REQUIREMENTS

The following are requirements for locating and managing excess RF Receive and Transmit cables of the MR systems sharing the Equipment Room.

- In the shared Equipment Room there must be > 6.6 feet (2 meters) separation between the RF receive cables of one system and the RF transmit cable of the other system. Refer to **Section 6 SYSTEM INTERCONNECTS** for Receive and Transmit cable Run Numbers.
- There must be > 6.6 feet (2 meters) separation between the system interconnect cables of each system sharing the Equipment Room space.
- Transmit cables in the Equipment Room must be cut to length to minimize excess cable length reducing the potential for signal coupling with other cables. No excess transmit cable can be stored in the Equipment Room.
- Receive cables excess length must be stored in a "figure 8" with overall dimension of < 3.3 feet (1 meter), see Illustration C-3.



PROPER STORAGE OF EXCESS RECEIVE CABLES
ILLUSTRATION C-3